

Euclid’s Elements — Book I

Proposition 39

Formal Reconstruction — Technical Documentation

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Equal triangles which are on the same base and on the same side are also in the same parallels.

1 Introduction

Proposition I.39 is the first genuine converse theorem in Euclid’s area block. Propositions I.35–I.38 proceed in the forward direction: geometric information about bases and parallels is converted into equality of content. Proposition I.39 reverses that direction. Two triangles on the same base are assumed to be equal, and from this equality one must recover a geometric conclusion about the position of their third vertices.

In the present formalization, Euclid’s equality of the two triangles is represented by the relation

```
HilbertScissorsEquicomplementable
```

Thus the theorem starts from a synthetic content relation rather than from numerical areas. The conclusion is also geometric rather than numerical: the two top vertices lie on a common Hilbert incidence line disjoint from the base line.

This converse direction exposes a new foundational issue. The formal scissors calculus developed for I.35–I.38 supports congruent replacement, splitting, addition, and equicomplementability, but it does not by itself express the strict statement that a whole figure cannot have the same content as one of its proper parts. Hartshorne isolates exactly this issue as property (Z), and he notes that Euclid’s proof of I.39 uses the principle that the whole is greater than the part.

The reconstruction therefore does not import a numerical area function and does not assume Hilbert’s Theorem 48 wholesale. Instead it isolates a configuration-free positivity principle for the scissors calculus: adjoining a nondegenerate triangle to an arbitrary scissors term cannot leave the content unchanged. A triangle split along an interior point of one side then derives, as a theorem, the specialized De Zolt statement actually used by I.39.

The result is a useful architectural checkpoint. The geometric part of the proof is derived from the existing Hilbert theory and Proposition I.37. The specialized proper-part statement is no longer an axiom. The remaining new foundational debt is the single positivity assumption `hilbert_scissors_triangle_positive`.

2 The Classical Configuration

Let ABC and DBC be triangles on the common base BC . The top vertices A and D lie on the same side of the line carrying BC , and the two triangles are assumed equal in content. The conclusion is that A and D lie on a line parallel to the base.

The classical reductio draws through A a line parallel to BC . If AD is not already this parallel, the new line meets the carrier of BD in a point E . There are two strict configurations compatible with A and D lying on the same side of the base:

$$B - E - D, \quad \text{or} \quad B - D - E.$$

In the first case EBC is a proper subtriangle of DBC . In the second case DBC is a proper subtriangle of EBC . The third collinear order

$$E - B - D$$

would place E and D on opposite sides of the base line, because the open segment ED would meet the base at B . The formal proof therefore excludes this third order from SameSide/OppositeSide theory rather than from the drawing.

Figure 1 displays exactly these three stages. Panel (a) is the public configuration. Panels (b) and (c) are the two proper-part alternatives that remain after the auxiliary parallel through A has been constructed.

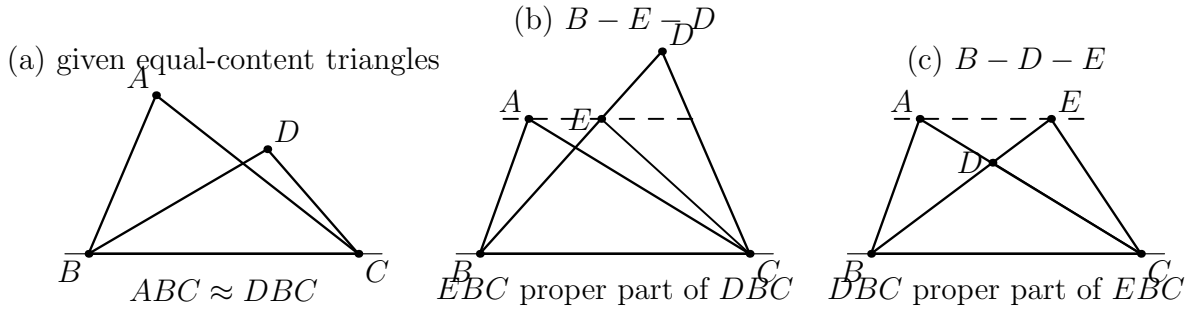


Figure 1: The geometric configuration used in the reconstruction of I.39. (a) The given triangles ABC and DBC share the base BC , lie on the same side of its carrier, and are assumed equicomplementable. (b) The parallel through A meets BD at E with $B - E - D$; Proposition I.37 gives equal content for ABC and EBC , while EBC is a proper subtriangle of DBC . (c) The alternative order is $B - D - E$; now DBC is a proper subtriangle of EBC . The order $E - B - D$ is excluded formally because it would put E and D on opposite sides of the base.

3 Classical Proof

Wilkins emphasizes that a complete proof must treat two positional configurations, although Euclid explicitly discusses only one of them. In each configuration the same auxiliary idea is

used.

Draw through A a straight line parallel to the common base BC . If this line already contains D , the conclusion is immediate. Otherwise let E be its intersection with the straight line BD , or with the continuation of BD beyond D .

The auxiliary triangle EBC has the same base BC as ABC , and its third vertex E lies on the parallel through A . Hence an earlier equal-content result shows that

$$ABC = EBC$$

in Euclid's area language. Wilkins' Study Note labels this step as Proposition I.36. The standard Book I dependency, and the Euclid/Joyce dependency table for I.39, identify the relevant triangle theorem as I.37: triangles on the same base and in the same parallels are equal. The present Lean development uses Proposition I.37 directly.

Now compare EBC with DBC . If

$$B - E - D,$$

then EBC is properly contained in DBC . If

$$B - D - E,$$

then DBC is properly contained in EBC . In either case the hypothesis that the original triangles have equal content, together with the equality $ABC = EBC$, would force a whole triangle to be equal to one of its proper parts.

This contradicts Euclid's Common Notion 5, that the whole is greater than the part. Therefore the assumption that D does not lie on the parallel through A is impossible. Hence the top vertices A and D lie on one straight line parallel to the common base.

The classical dependency is therefore schematically

$$\begin{array}{c}
\text{construct the parallel through } A \\
\Downarrow \\
E \in BD \text{ or its extension} \\
\Downarrow \\
ABC = EBC \text{ by an earlier equal-content theorem} \\
\Downarrow \\
EBC \text{ is a proper part of } DBC \text{ or conversely} \\
\Downarrow \text{ C.N.5} \\
\text{contradiction} \\
\Downarrow \\
AD \parallel BC.
\end{array}$$

The point of the reconstruction is not to suppress this argument but to make its hidden logical requirement explicit.

4 Property (Z), De Zolt, and the Converse Content Problem

4.1 Why I.39 is different from I.38

Hartshorne's analysis of Euclid's area theory makes the boundary especially clear. Once an adequate equal-content relation is available, I.38 follows by transitivity. I.39 is its converse,

and the converse uses the claim that a proper part cannot have the same content as the whole. Hartshorne calls the relevant strictness condition property (Z).

Thus the implication

$$\text{same parallels} \longrightarrow \text{equal content}$$

belongs to the forward scissors theory, while

$$\text{equal content} \longrightarrow \text{same parallels}$$

requires an additional strictness principle.

This is exactly what the current formalization detects. The operations already present in `HilbertScissors` do not yield a contradiction merely from the fact that one triangle term arises from a geometric proper subtriangle of another. The term calculus intentionally has no primitive notion of geometric interior or nonoverlap.

4.2 Hilbert's Theorem 48

Hilbert reaches the converse by a different route. In Chapter IV he first develops segment arithmetic and an area function. In Section 21, equicomplementable polygons are shown to have equal areas. For two equicomplementable triangles with equal base g and altitudes h, h' , Hilbert writes the equality

$$\frac{1}{2}gh = \frac{1}{2}gh'$$

and cancels the common nonzero factor to obtain

$$h = h'.$$

This is Theorem 48.

That theorem would certainly be strong enough for I.39. It is not, however, the theorem formalized here. Importing it directly would hide the exact foundational obstacle encountered by the present scissors language and would force the development prematurely into multiplication of segment classes and the later area theory.

4.3 Positivity and the derived De Zolt principle

Hilbert later states a De Zolt theorem asserting, in effect, that after a rectilinear figure has been decomposed into triangular pieces, omitting one piece cannot leave enough material to reconstruct the whole. The current formalization isolates the strictness behind that phenomenon at a more general algebraic boundary.

The one new temporary content assumption is:

```
axiom hilbert_scissors_triangle_positive
  [HilbertIncidence Geo]
  (P : HilbertScissorsTerm Geo)
  (A B C : Geo.Point)
  (hNoncol : Not (Collinear Geo A B C)) :
  Not (HilbertScissorsEquicomplementable Geo
    (P + hilbertScissorsTriangle Geo A B C)
    P)
```

This statement contains no betweenness, parallelism, or proposition-specific configuration. It says only that adjoining a nondegenerate triangular piece strictly changes content.

The geometric proper-part statement needed in I.39 is then derived. First,

```
scissors_split_side
```

uses the primitive scissors split to prove that, when $B - X - A$,

$$ABC \sim XBC + AXC,$$

where $\sim$ denotes `HilbertScissorsEq`. The proof also establishes that the cut-off triangle AXC is nondegenerate. If one assumed

$$ABC \approx XBC,$$

then transport across the split would give

$$XBC + AXC \approx XBC,$$

contradicting positivity.

Consequently the former I.39 axiom is now a theorem:

```
theorem hilbert_scissors_triangle_proper_part
  [HilbertIncidence Geo]
  [HilbertOrder Geo]
  (A X B C : Geo.Point)
  (hBXA : Geo.Between B X A)
  (hBC : Ne B C)
  (hNoncol : Not (Collinear Geo B C A)) :
  Not (HilbertScissorsEquicomplementable Geo
    (hilbertScissorsTriangle Geo A B C)
    (hilbertScissorsTriangle Geo X B C))
```

The argument should not be read as a proof that positivity is equivalent, in the present Lean API, to Hilbert's Theorem 48. What has been established formally is narrower and cleaner: positivity implies the specialized proper-part theorem, and that theorem is sufficient for I.39. A future concrete additive valuation, nonzero on nondegenerate triangles, will discharge the positivity axiom directly.

5 Proof Architecture of the Reconstruction

The Lean proof separates the geometry from the one missing content principle.

5.1 Step 1: the coincident-vertex boundary case

If

$$A = D,$$

there is no need for a reductio. The point A lies off the base because the `SameSide` hypothesis contains this information. The existing parallel-through- point theorem constructs a line through A disjoint from the base, and the same incidence fact is used for both top vertices.

This boundary case is invisible in the usual diagram but must be handled explicitly because the strict point-line parallel relation requires two distinct points.

5.2 Step 2: the already-parallel branch

Assume now that $A \neq D$. The proof tests whether

$$AD \parallel BC.$$

If so, the helper

```
proposition39_parallel_points_to_top_line
```

converts point-line parallelism into the incidence-line form used by the public conclusion. It constructs the line through A, D and proves that it is disjoint from the given incidence line **base** through B, C .

5.3 Step 3: construction of the auxiliary point

The only nontrivial branch is

$$A \neq D, \quad AD \not\parallel BC.$$

The helper

```
proposition39_auxiliary_point
```

draws the parallel through A to BC and proves that the carrier of BD must meet it. If the two incidence lines were disjoint, Euclidean uniqueness of the parallel through B would force the carrier BD to coincide with the base, contradicting the fact that D lies off the base.

The resulting point E satisfies

$$\begin{aligned} &B, E, D \text{ collinear,} \\ &E \neq B, \quad E \neq D, \\ &E \notin \text{base,} \\ &\text{SameSide}(A, E; \text{base}), \end{aligned}$$

and either

$$E = A$$

or

$$AE \parallel BC.$$

The last disjunction is exactly what is needed for Proposition I.37. If $E = A$, the content comparison is reflexive; otherwise the nondegenerate parallel $AE \parallel BC$ gives it directly.

5.4 Step 4: normalization of the order on BD

The helper

takes the collinearity of B, E, D and proves

$$B - E - D \quad \text{or} \quad B - D - E.$$

The raw betweenness trichotomy has a third branch, $E - B - D$. The proof excludes it by side-of-line theory. Because B lies on the base, $E - B - D$ would make E and D opposite sides of the base. But transitivity and symmetry of SameSide, using the common point A , show that E and D are on the same side. The two statements are incompatible.

This is one of the main pieces of hidden configuration data revealed by the formalization.

5.5 Step 5: the I.37 content comparison

The auxiliary parallel gives

$$ABC \approx EBC,$$

where $\approx$ denotes HilbertScissorsEquicomplementable. The original hypothesis gives

$$ABC \approx DBC.$$

Symmetry and transitivity therefore yield

$$DBC \approx EBC.$$

No subtraction, cancellation, numerical area, or new scissors constructor is used here. This part is entirely inside the content interface already established before I.39.

5.6 Step 6: the proper-part contradiction

There are now exactly two order branches.

If

$$B - E - D,$$

then EBC is a proper subtriangle of the nondegenerate triangle DBC . The derived proper-part theorem says

$$DBC \not\approx EBC,$$

contradicting the comparison obtained above.

If

$$B - D - E,$$

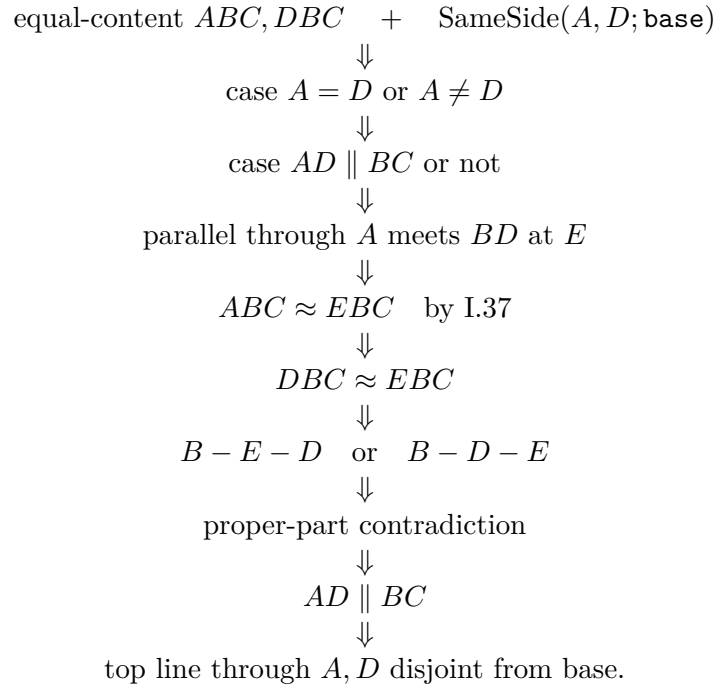
then DBC is a proper subtriangle of the nondegenerate triangle EBC . After symmetry of equicomplementability, the same derived theorem gives the opposite contradiction.

Thus the assumption

$$AD \nparallel BC$$

has no possible realization. The remaining branch is $AD \parallel BC$, which is converted into the required incidence line through A and D disjoint from the base.

The formal DAG can be summarized as



6 Lean Formalization

The current file imports

```
import CGJteamLab.Proposition37
import CGJteamLab.HilbertScissorsPositivity
```

The positivity module in turn imports `HilbertScissorsInvariant`, which contains the generic valuation interface intended for the eventual discharge of the positivity assumption.

The public theorem is:

```
theorem euclid_proposition_39
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]
  (A B C D : Geo.Point)
  (base : Geo.Line)
  (hBC : Ne B C)
  (hBbase : HilbertIncidence.OnLine B base)
  (hCbase : HilbertIncidence.OnLine C base)
  (hSame : HilbertSameSide Geo A D base)
  (hEqual :
    HilbertScissorsEquicomplementtable
      Geo
      (hilbertScissorsTriangle Geo A B C)
      (hilbertScissorsTriangle Geo D B C)) :
  Exists fun top : Geo.Line =>
    HilbertIncidence.OnLine A top /\
    HilbertIncidence.OnLine D top /\
    HilbertLinesDisjoint Geo top base := by
  ...
```


The ellipsis is documentation notation only. The repository theorem is fully proved and contains no `sorry`. Unicode logical symbols from the source signature have been normalized above to ASCII forms for the documentation listing; declaration names and argument structure are unchanged.

The mathematically significant proposition-local declarations are

```
proposition39_auxiliary_point
proposition39_auxiliary_order
proposition39_parallel_points_to_top_line
```

The content module contributes the chain

```
scissors_split_side
hilbert_scissors_triangle_positive
hilbert_scissors_triangle_proper_part
```

where the middle declaration is the axiom and the last declaration is a theorem derived from the first two.

The final axiom audit reported by Lean is:

```
'Geometry.euclid_proposition_39' depends on axioms:
[propext,
 Classical.choice,
 bookZero_nullSegment1,
 hilbert_scissors_triangle_positive,
 Quot.sound]
```

Here `propext`, `Classical.choice`, and `Quot.sound` are logical foundations inherited by the Lean development. The pre-existing Book Zero debt shown in the audit is inherited transitively. The only new I.39-specific content assumption is the positivity declaration shown there.

The full project command

```
lake build
```

is clean with the current `Proposition39.lean`.

7 Relationship with Earlier Propositions

7.1 Proposition I.31

Classically, the proof requires drawing through A a line parallel to the base. Euclid would cite the parallel construction of I.31. The Lean proof does not call I.31 by name. The construction is already available as

```
hilbert_parallel_through_point_exists
```

in the developed Hilbert interface.

This is another example of the distinction between the historical proposition DAG and the actual dependency graph of the reconstruction.

7.2 Proposition I.37

I.37 is the decisive earlier proposition in the actual Lean proof. Once E has been constructed on the parallel through A , the two triangles ABC and EBC have the same base and lie between the same parallels. Therefore

$$ABC \approx EBC.$$

This use of I.37 is particularly important because the project’s proof of I.37 was deliberately constructed without a general “halves of equals” or property- (Z) principle. Consequently there is no circularity: I.37 provides the forward equal-content theorem, while I.39 is exactly the later converse that first introduces a strict proper-part assumption.

7.3 Proposition I.38

I.38 is the immediate conceptual contrast. Its content conclusion is still forward and can be obtained from a congruent-copy construction, I.37, and transitivity. No strict comparison of contents is needed.

I.39 starts with equal content and concludes a geometric relation. This is why the existing I.35–I.38 scissors interface is almost sufficient but not quite: the missing ingredient is not another equality operation but a strictness principle.

7.4 Proposition I.40

The clean project build confirms that the public theorem name `euclid_proposition_39` is now available to Proposition I.40. Thus I.39 is not an isolated experimental theorem; it is part of the continuing Book I dependency chain.

8 Hilbert and Book Zero Components Used

8.1 Incidence and order

The auxiliary construction uses the Hilbert incidence API for:

- the line through two distinct points;
- uniqueness of a line through two distinct points;
- conversion between point-line carriers and incidence lines;
- collinearity induced by a common incidence line;

- betweenness trichotomy for three distinct collinear points;
- transport of betweenness to incidence information.

These tools are responsible for proving that the constructed point E is actually on the carrier BD and for reducing its position to the two strict orders needed by the reductio.

8.2 SameSide and OppositeSide

The public hypothesis

$$\text{SameSide}(A, D; \text{base})$$

contains more information than a conventional drawing usually records. It implies that both top vertices lie off the base. After the auxiliary point is constructed, the proof establishes

$$\text{SameSide}(A, E; \text{base})$$

and hence, by symmetry and transitivity,

$$\text{SameSide}(E, D; \text{base}).$$

This is used against

```
hilbert_oppositeSide_not_sameSide
```

to eliminate the order $E - B - D$.

8.3 Euclidean parallelism

The reconstruction is Euclidean in the current project API. Parallelism is used in two logically distinct ways.

First, the proof constructs a parallel through A to the base. Second, Euclidean uniqueness is used in the contradiction showing that the carrier BD cannot be disjoint from this new parallel unless it coincides with the base, which would place D on the base.

The helper

```
collinear_parallel_trans
```

then transports the auxiliary parallel along a collinear carrier so that, when $E \neq A$, the exact hypothesis

$$AE \parallel BC$$

required by I.37 is available.

8.4 Scissors content

The content stage uses only the already established equivalence operations:

- `equicomplementable_refl`;
- `equicomplementable_symm`;
- `equicomplementable_trans`;
- Proposition I.37.

No new scissors constructor is introduced. The existing split constructor is packaged by `scissors_split_side`. The specialized proper-part principle is derived from that split and the single new positivity axiom.

8.5 Book Zero

No new Book Zero theorem is introduced for I.39. The axiom audit nevertheless shows the inherited dependency

```
bookZero_nullSegment1
```

through earlier project infrastructure. This dependency predates I.39 and is not part of the new converse-content argument.

9 Formalization Notes

9.1 Geometry, representation, and content

From I.35 onward it is important to keep three layers separate.

Geometry. The proof constructs the parallel through A , proves that it meets the carrier BD , obtains the point E , establishes its side-of-line position, classifies the order of B, E, D , and finally proves the required parallelism.

Representation. The public theorem speaks about an explicit incidence line `base` and returns an explicit incidence line `top`. Proposition I.37, in contrast, uses point-line parallelism. The dedicated top-line packaging helper bridges these two representations. Its exact declaration name is

```
proposition39_parallel_points_to_top_line
```

This is a representation issue, not an additional geometric axiom.

At the triangle-term level no special permutation machinery is central to I.39: all three triangles are written on the common ordered base BC .

Content. The equality information is entirely synthetic. The formal theorem never assigns a number to the area of a triangle. It derives

$$DBC \approx EBC$$

by symmetry and transitivity of equicomplementability and then invokes one strict-content principle to contradict the proper inclusion.

Thus the proof does not identify `HilbertScissorsEquicomplementable` with numerical equality of area. It uses only the operations explicitly available in the formal content layer.

9.2 Hidden configuration data

The Lean proof establishes explicitly:

- A and D are off the base line;
- $B \neq D$ because B is on the base and D is off it;
- the parallel through A is disjoint from the base;
- the carrier BD must meet that parallel in the nonparallel branch;
- the intersection point E is off the base;
- $E \neq B$ and $E \neq D$;
- B, E, D are collinear;
- A and E are on the same side of the base;
- therefore E and D are on the same side of the base;
- the order $E - B - D$ is impossible;
- in the remaining two orders the larger triangle is nondegenerate;
- the proper-subtriangle relation is therefore strict rather than degenerate.

None of these facts is read from Figure 1. The diagram is a summary of conclusions proved inside the geometry layer.

9.3 Axiomatic status

The public theorem assumes

`[HilbertEuclideanPlane Geo].`

The proof uses incidence, order, congruence-derived side-of-line theory, and Euclidean parallelism. It introduces no continuity axiom and no Archimedean axiom.

There is, however, one new non-geometric assumption:

```
hilbert_scissors_triangle_positive
```

This should be read as an explicit positivity axiom for content, not as part of Hilbert Group IV. Its role is analogous to the strictness isolated by Hartshorne’s property (Z) and to the De Zolt phenomenon in the classical theory of area. The proposition-specific proper-part declaration is now a theorem:

The current theorem is therefore conditional in a very precise sense: all geometry of I.39 is proved from the established Hilbert/Euclidean infrastructure, while the missing semantic strength of the term-level scissors calculus is exposed as one configuration-free positivity assumption.

9.4 Why Hilbert Theorem 48 is not an imported axiom

An earlier experimental route packaged the converse as an equal-altitude statement close to Hilbert Theorem 48. That route was abandoned for the final I.39 module because it concealed too much of the real dependency.

Hilbert’s own proof of Theorem 48 is downstream from segment arithmetic and the area function. Reproducing it faithfully would require substantially more infrastructure than I.39 itself needs. The present route instead follows the classical *reductio* and isolates the exact strict-content principle at the point where the current formal scissors calculus stops.

This makes the remaining debt cleaner and easier to study independently. The module `HilbertScissorsInvariant` already provides the generic valuation interface and proves that equicomplementable terms have equal valuation. What is still missing is a concrete additive valuation whose value on every nondegenerate triangle is nonzero. Such a valuation immediately discharges `hilbert_scissors_triangle_positive`, after which the derived proper-part theorem and I.39 require no change.

9.5 Source audit

The sources play different roles in the reconstruction.

Wilkins supplies the clearest complete audit of the classical configuration. His Study Note explicitly distinguishes the two positions of the intersection point E and notes that Euclid discusses only one of them. This two-case analysis matches the two strict betweenness branches retained by the Lean proof.

Hartshorne supplies the semantic diagnosis. Section 22 distinguishes forward equal-content arguments from converse arguments and explicitly states that I.39 uses “the whole is greater than the part,” hence property (Z). This is the principal source for interpreting the new formal positivity requirement.

Hilbert supplies the broader theory of plane area. Theorems 44–46 justify equicomplementability as the correct synthetic target for the forward equal-base/equal-altitude theorems. Section 21 proves Theorem 48 only after the development of area, while Theorem 52 records a De Zolt-type impossibility of reconstructing a whole after omitting a triangular part. These results explain both why the converse requires strictness and why it is reasonable to isolate a positivity principle from which the proper-part statement can be derived.

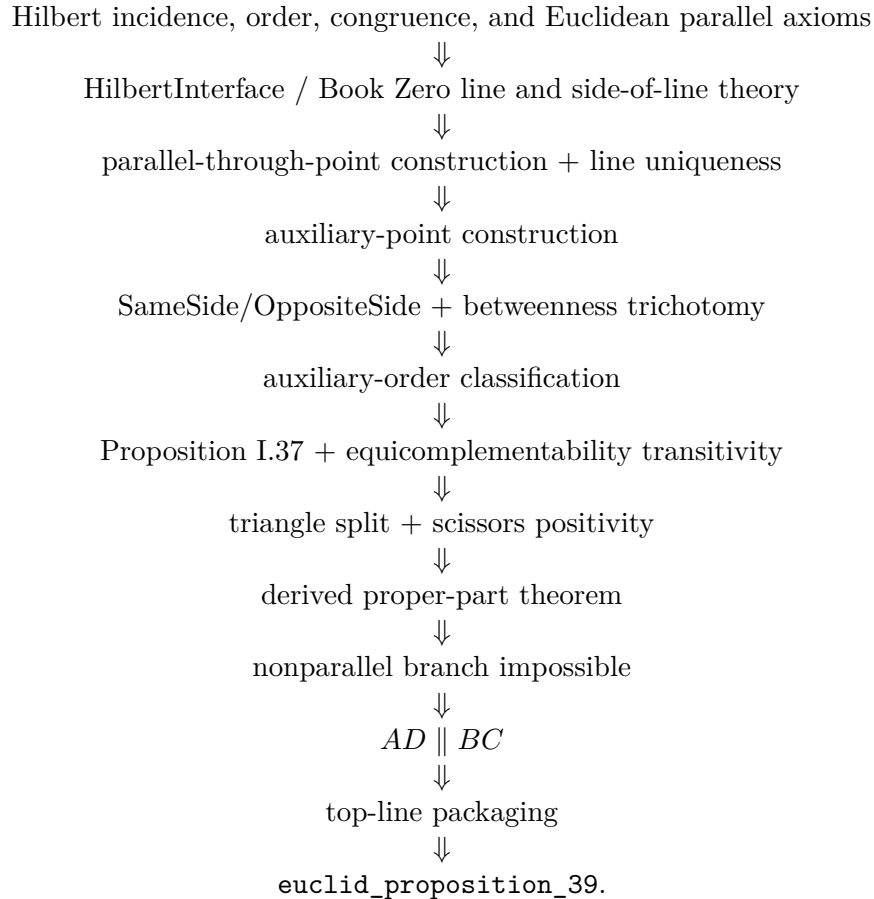
The Beeson proof corpus remains a control source for hidden positional and order obligations, but it is not used as the architecture of the final Lean proof. In particular, the project does not import a Tarski-style proof trace or reorganize the Hilbert interface around Beeson tactics.

Borsuk–Szmielw and Greenberg remain control sources for incidence, order, parallelism, and the neutral/Euclidean boundary. They are not used here to replace the content relation by a numerical area theory.

The Euclid/Joyce edition controls the classical statement and Common Notion 5. The project Book I compilation is used as a consistency check against the documentation of I.35–I.38. Neither source overrides the current Lean module, which is authoritative for the formal dependency graph.

10 Dependency Summary

The actual formal dependency chain is



The historical and formal DAGs are therefore related but not identical:

Euclid/Wilkins:	parallel through $A \rightarrow E \rightarrow$ earlier equal-area theorem $\rightarrow$ C.N.5,
Lean:	parallel through $A \rightarrow E \rightarrow$ I.37 $\rightarrow$ derived proper-part theorem.

New geometric axiom:	no
New content axiom:	yes — scissors positivity
New proposition-local helpers:	yes
New reusable Scissors theorems:	yes — split-side and proper-part
New Book Zero theorem:	no
New numerical area or polygon semantics:	no
Public theorem compiles:	yes
Full <code>lake build</code> :	clean
<code>sorry</code> in Proposition39.lean:	no

11 Conclusion

Proposition I.39 marks a real change in the logical direction of Euclid’s area theory. The preceding propositions show that appropriate geometric configurations imply equality of content. I.39 asks equal content to force a geometric conclusion. That reversal is exactly where a strict content principle becomes unavoidable.

The formal reconstruction makes the boundary unusually transparent. All of the geometry needed for the classical reductio – the parallel through A , its intersection with BD , the SameSide analysis, the collinear order trichotomy, and the reduction to Proposition I.37 – is derived inside the existing Hilbert/Euclidean infrastructure. No new geometric axiom is needed.

What remains is one explicit content assumption: adjoining a nondegenerate triangle to an arbitrary scissors term cannot leave the content unchanged. Rather than encoding the I.39 configuration itself as an axiom, the development records this configuration-free positivity statement in the declaration

```
hilbert_scissors_triangle_positive
```

The triangle split then proves the proposition-specific proper-part statement:

```
hilbert_scissors_triangle_proper_part
```

The axiom audit confirms that positivity, not the proper-part statement, is the only new I.39-specific foundational debt.

This is the main structural result of the proposition. I.39 itself is now stable. A future positive additive valuation can discharge the remaining positivity assumption without reopening either the proper-part derivation or the geometric proof of I.39.